

THE DIGESTIVE SYSTEM

INTRODUCTION

The digestive system is used for breaking down food into nutrients which then pass into the circulatory system and are taken to where they are needed in the body. There are four stages to food processing -

1. Ingestion - taking the food
2. Digestion - breaking down food into nutrients
3. Absorption - taking in nutrients by cells
4. Egestion - removing any leftover wastes

ORGANS OF DIGESTIVE SYSTEM

- Digestive tract
- 3 pairs of salivary glands
- Liver, gall bladder and pancreas

Digestive tract is composed of oral cavity, pharynx, oesophagus, stomach, duodenum, jejunum, ileum, caecum, ascending colon, transverse colon, descending colon, sigmoid colon, rectum and anal canal.

GUT WALL

It consists of four layers -

1. Outer serosal layer
2. Smooth muscle layer which consists of outer longitudinal and inner circular layer. Contraction of the circular layer leads to the narrowing of the lumen and that of longitudinal muscle leads to shortening in length.
3. Submucous layer
4. Mucous layer

NERVE SUPPLY OF THE GI TRACT

1. Extrinsic nerve supply - sympathetic (inhibitory) system and parasympathetic (stimulatory) system.
2. Intrinsic nerve supply - myenteric plexus (Auerbach plexus) between circular and longitudinal muscle layer. Submucosal plexus (Meissner's plexus) in submucous layer

ORAL CAVITY (MOUTH)

Structures in the mouth aid in digestion are -

- TEETH - cut, tear, crush and grind food
- SALIVARY GLANDS - salivary gland contains serous cells and mucous cells. Serous cells secrete ptyalin and mucous cells secrete mucin.
 - MAJOR SALIVARY GLANDS
 - Parotid
 - Submandibular
 - Sublingual
 - Minor salivary gland
 - Lingual mucous
 - Lingual serous
 - Buccal glands

- Labial glands
- Palatal glands

SALIVARY SECRETIONS

- Composition of saliva - 1500ml/day
 - o Enzymes - ptyalin or salivary alpha amylase, kallikrein, lysozyme, lipase
 - o Mucin
 - o Ions - Na⁺, K⁺, Ca²⁺, Cl⁻, HCO₃⁻, phosphate
 - o pH- below 7
- function of saliva
 - o ptyalin - breaks down starch to maltose
 - o mucin - lubricate the food, thus assist mastication & facilitates to swallow
 - o lingual lipase - helps in triglyceride digestion
 - o it keeps the mouth moist and act as solvent for molecules that stimulates taste buds
 - o reduces the risk of buccal infection and dental caries as it contains lysozyme which kills bacteria

MASTICATION (CHEWING)

Mechanical process by which larger food breaks up into smaller pieces which -

- makes it easier for the food to be swallowed
- mixes food with salivary secretions

muscles of mastication includes - masseter, temporalis, buccinator, internal and external pterygoid.

DEGLUTITION (SWALLOWING)

Reflex response controlled via vagus nerve. It occurs in three stages -

1. stage 1: oral stage - voluntary stage
2. stage 2: pharyngeal stage - involuntary stage
3. stage 3: oesophageal stage - involuntary stage

1. ORAL STAGE

- a. After mastication the chewed food is rolled into bolus by movements of cheeks and tongue
- b. Swallowing starts by closure of the mouth & voluntary contraction of mylohyoid muscle which throws the bolus back into posterior pharyngeal wall. This constitute deglutition reflex.

2. PHARYNGEAL STAGE (1-2 SEC)

- a. Pushes the bolus into oesophagus
- b. Here soft palate is elevated to close off the nasal cavity
- c. Vocal cords are approximated & breathing is momentarily stopped (deglutition apnea). Epiglottis guards the laryngeal opening until bolus reaches oesophagus. This prevents aspiration of food into larynx.

3. OESOPHAGEAL STAGE (10-15 SEC)

- a. Oesophagus separated from the oral cavity by upper oesophageal sphincter and from stomach by lower oesophageal sphincter (cardiac sphincter)

- b. Upper oesophageal sphincter opens reflexly upon swallowing & allows bolus to enter into the body of oesophagus.
- c. Peristalsis begins which propels the food into stomach. Distension of oesophagus causes reflex relaxation of lower oesophageal sphincter which allows bolus to enter into stomach.

DISORDERS OF SWALLOWING

1. Achalasia cardia - characterised by the failure of lower oesophageal sphincter to relax completely on swallowing due to destruction of local nerve plexuses. Constriction of sphincter causes food accumulation in the oesophagus and proximal oesophagus dilates.
2. Dysphagia - difficulty in swallowing. It may occur due to disorder involving any stage of swallowing.

HORMONES OF THE GASTROINTESTINAL TRACT

The word hormone is derived from the Greek word 'HORMEIN', which means to execute or to arouse. It is secreted from the ductless glands, goes to the circulation and acts on their target cells, where they elicit physiologic, morphologic and biochemical responses.

THE MAIN FUNCTION OF THE GASTROINTESTINAL TRACT ARE -

- Digestion
- Absorption
- Nutrition

CLASSIFICATION

- Gastrin family
 - Gastrin
 - Cholecystokinin
- Secretin family
 - Secretin
 - Gastric inhibitory peptide
 - Vasoactive intestinal peptide
 - Glucagon
 - Insulin
 - Leptin

GASTRIN

- Secreted by 'G' cells, located in the pyloric region of the stomach
- It is flask shaped, secreted in inactive form, converted to active form by HCL.
- Three types are there - G-34, G-17, G-14
- Its plasma half-life is 2-3 minutes
- Inactivated mainly in the kidney and the small intestine

FUNCTIONS -

- Stimulates HCL secretion
- Increases intestinal and gastric motility
- Increases pancreatic secretion of INSULIN and GLUCAGON
- TROPHIC ACTION i.e. maintain the health of gastric mucous membrane.

FACTORS AFFECTING SECRETIONS

- Vagal stimulation
- Distension of the pyloric antrum
- Products of protein digestion
- Calcium and epinephrine

FACTORS INHIBITING SECRETION

- Low pH of gastric juice (less than 3.)
- Somatostatin released by D cells.
- Others - SECRETIN, GIP, VIP, GLUCAGON, CALCITONIN

APPLIED

- Zollinger ellison syndrome
- Some pancreatic tumors of delta cells, secrete excess HCL and causes peptic ulceration
- Gastrinoma

CHOLECYSTOKININ PANCREOZYMIN

- It is secreted from the I cells of the duodenum and jejunum.
- It is a polypeptide containing 33 amino acids

ACTIONS

- Contraction of gall bladder to release bile.
- Potentiates the secretion of 'SECRETIN' to produce more alkaline juice.
- Inhibits the GASTRIC MOTILITY.
- Increases the motility of small and large gut.
- Trophic action
- It is also involved in regulation of food intake

APPLIED

- Acute pancreatitis
- Chronic pancreatitis

SECRETIN

- It was the first hormone to be discovered in 1902, by BAYLISS and STARLING.
- Secreted by 'S' cells of duodenum and jejunum.
- It is a polypeptide with 27 amino acids.
- Also stimulates 'BILE' secretion.
- Potentiates the effect of CCK on pancreas.
- Causes contraction of the pyloric sphincter delaying gastric emptying and preventing duodenal reflux.

GLUCAGON

- It is secreted by alpha cells of the 'ISLETS OF LANGERHANS' when the blood glucose level falls.
- It is a polypeptide containing 29 amino acids
- It is also secreted in the 'L' cells in the lower and also in the brain.

INCREASE SECRETION

- Hypoglycemia
- Stimulation of the sympathetic nerves to the pancreas
- Vagal stimulation
- Protein meal
- During starvation
- CCK
- Gastrin

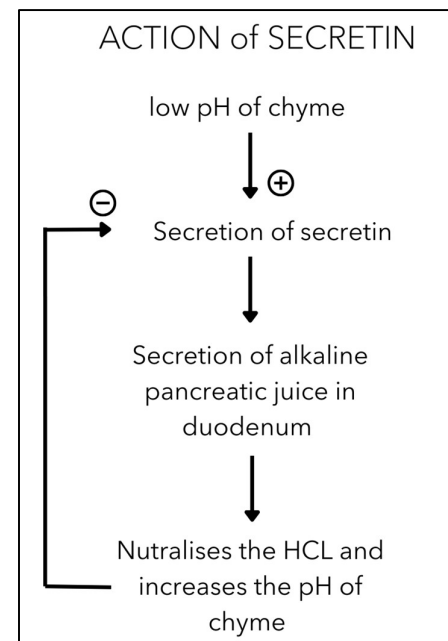
DECREASE SECRETION

- Secretin
- Rise in plasma glucose level

ACTIONS

ON CARBOHYDRATE METABOLISM

- Increase glycogenolysis in the liver.
- Increase gluconeogenesis FROM lactate, pyruvate, glycerol, amino acids



INSULIN

- It is a peptide hormone, released by beta cells of the pancreas
- Discovery by BENTING and BEST. In 1922.
- It is secreted when the blood glucose exceeds the normal level i.e. 70-110 mg/dl
- Insulin receptors are present in most cells, [INSULIN DEPENDENT]
- Cells like - BRAIN, KIDNEY, LINING OF G.I.T, RBC CELLS [INSULIN INDEPENDENT] .
no insulin receptors can absorb GLUCOSE without insulin.
- EFFECT OF INSULIN ON VARIOUS TISSUE
 - Adipose tissue
 - Muscle
 - Liver
 - General
- APPLIED
 - Diabetes mellitus
 - It is a syndrome of impaired carbohydrate, fat and protein metabolism caused by either lack of insulin secretion or sensitivity of the tissue to insulin.
 - Types - TYPE -1 , TYPE- 2

THE STOMACH

- The stomach is dilated part of the alimentary canal that between oesophagus and small intestine.
- It is a muscular sac
- It is J shaped.

THE STOMACH IS DIVIDED INTO FOUR REGIONS

1. The **cardia**, which surrounds the opening of the esophagus into the stomach.
2. The **fundus of stomach**, which is the area above the level of the cardiac orifice.
3. The **body of stomach**, which is the largest region of the stomach.
4. The **pyloric part**, which is divided into the **pyloric antrum** and **pyloric canal** and is the distal end of the stomach.

LINING OF GASTRIC MUCOSA

- Inner surface of stomach is highly folded into rugae.
- Gastric mucosa has gastric pits.
- Cells that lines the folds deeper in the mucosa are the gastric glands
- Gastric glands are situated within the mucous membrane in groups of 2-7 which together communicate with the gastric lumen through gastric pits.

GASTRIC GLANDS ARE MAINLY THREE TYPES

1. Main gastric glands - found in body and fundus of stomach. Cells of these glands are 2 types
 - a. Chief/peptic cells - secrete pepsinogen
 - b. Parietal/oxyntic cells - secrete hydrochloric acid
2. Cardiac tubular glands - secrete soluble mucous
3. Antral gland - contain 2 types of cells
 - a. Mucus cells - secrete soluble mucus

- b. G cells – secrete Gastrin

CHIEF CELLS/ PEPTIC CELLS

Situated in the wall of the glands near the base, Secretes pepsinogen.

PEPSINOGEN –

The functions of pepsinogen are –

- a. Pepsinogen digest proteins to polypeptides by hydrolysing peptide bonds. pH for this action requires <4.0
- b. Pepsin acts on water soluble caseinogen (milk protein) to form soluble casein. This combines with calcium salts to form insoluble calcium caseinate which is digested enzymatically.

PARIETAL CELLS/OXYNTIC CELLS

It is found in the body of the stomach. They secrete HCL as well as INTRINSIC FACTOR (IF). pH of this fluid is 1.0

FUNCTION OF HCL

- 1. Provides H⁺ concentration necessary for pepsin activity.
- 2. Activates pepsinogen to pepsin.
- 3. Kills ingested bacteria
- 4. Helps in iron absorption by converting ferric to ferrous form.

*The parietal cells also secrete IF which combines with vitamin B12 and helps in absorption of the latter from the small intestine.

CARDIAC TUBULAR GLAND

Secretes soluble mucus.

PYLORIC GLAND/ANTRAL GLANDS

They are found in the mucosa between the level of incisura angularis on lesser curvature and pylorus. Mucus cells secrete mucus. Deeper part of the pyloric gland have G cells which secrete **GASTRIN**

FUNCTION OF STOMACH

- 1. Storage
- 2. Digestive
- 3. Function of HCL
- 4. IF (intrinsic factor) : requires for absorption of vit B12. Decrease absorption produces megaloblastic anemia.
- 5. Food released at a controlled rate into the duodenum to provide proper time for digestion and absorption by small intestine.

COMPOSITION OF GASTRIC JUICE

Daily secretion	:	2.5 – 3 liters/day, isotonic
Ph	:	1 – 2 (acidic)
Water	:	99.45%
Solids	:	0.55%

SOLIDS

- A. Electrolytes
 - a. Cation : Na^+ ; K^+ ; H^+ ; Mg^{2+}
 - b. Anions : Cl^- ; HCO_3^- ; HPO_4^{2-} ; SO_4^{2-}
- B. Enzymes
 - a. Pepsinogen
 - b. Gelatinase – liquifies gelatin
 - c. Gastric lipase
 - d. Lysozymes – Bacteriocidal
 - e. Urease – Hydrolyse urea to produce ammonia
- C. Mucus
 - a. Soluble mucus – pyloric and cardiac tubular glands
 - b. Visible mucus – surface epithelium of gastric mucosa
- D. Intrinsic factor – absorption of vitamin B12

MECHANISM OF HCL SECRETION

- 1. Secretion of H
 - a. Carbonic anhydrase helps in formation of H_2CO_3 from CO_2 and H_2O within the parietal cells
 - b. H_2CO_3 dissociates to form H^+ and HCO_3^-
 - c. H^+ is secreted into the lumen of canaliculi of parietal cell in exchange for K^+ by H^+-K^+ ATPase pump
 - d. HCO_3^- produced are transported out through antiport in exchange with Cl^- at basolateral membrane.
- 2. Secretion of Cl^-
 - Because of high intracellular negativity, Cl^- is forced out through Cl^- channel on apical membrane.
 - Cause of high intracellular negativity:
 - 1. Na^+-K^+ pump located at the basolateral membrane pumps out 3Na^+ for 2K^+
 - 2. K^+ pumped in diffuses out through K^+ channels present on basolateral as well as apical membrane.

REGULATION OF SECRETION OF GASTRIC JUICE

There are two regulatory mechanisms for regulation of secretion of gastric juice

- 1. Nervous mechanism –
 - a. Local autonomic reflexes
 - b. Impulses from the CNS  X nerve (vagal nerve)
- 2. Humoral mechanism
 - a. Mediated by gastrin and histamine

PHASES OF GASTRIC JUICE SECRETION

Neural and hormonal mechanisms regulate the release of gastric juice. Stimulatory and inhibitory events occur in 3 phases

- 1. Cephalic phase
- 2. Gastric phase
- 3. Intestinal phase

CEPHALIC PHASE	Gastric phase	Intestinal phase
• Psychic stimuli • Vagally mediated • 30-50% of total gastric juice secretion.	• Presence of food in the stomach • Local reflex and responses to gastrin • 50-60% of total gastric juice secretion	• Presence of food in the duodenum • Reflex and hormonal feedback • Very less secretion.

PEPTIC ULCER

It is the breakdown of the mucosal epithelium of stomach or duodenum

Causes -

1. Infection with a bacterium, helicobacter pylori. It stimulates H^+ secretion and inhibits HCO_3 secretion
2. Ethanol
3. Bile salts

GALL BLADDER

BILE

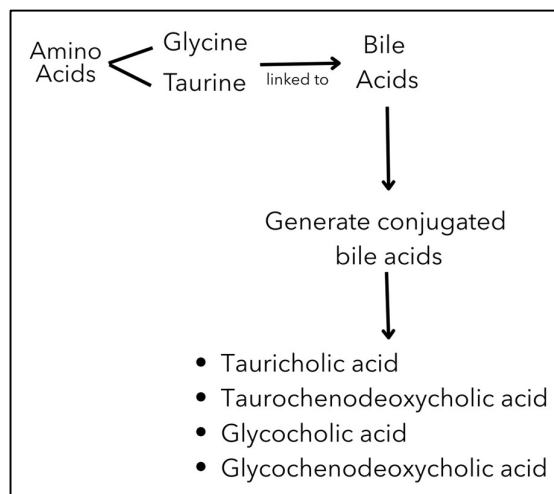
- Digestive juice formed in liver
- Hepatocytes to bile canaliculi
- Common hepatic duct
- Cystic duct
- Common bile duct

COMPOSITION OF BILE

- Bile formed in hepatocytes and released into canaliculi
- Its an alkaline juice comprised of -
 - Water
 - Solids
 - Organic- bile salts, pigments, cholesterol, lecithin, FA, alkaline phosphatase
 - Inorganic- Na^+ , K^+ , Ca^+ , HCO_3^- & Cl^-

FORMATION OF BILE SALTS

- Bile acids are formed from cholesterol
- Primary bile acids are- cholic acid and chenodeoxycholic acid
- Secondary bile acids - primary bile acids converted by colon bacteria to secondary bile acids i.e. deoxycholic acid and lithocholic acid
- Conjugation of bile acid with taurine and glycine occurs in liver. The conjugates tauro and glycocholic acids, from sodium and potassium salts, which are called bile salts.



ENTEROHEPATIC CIRCULATION OF BILE SALTS

- Of the total bile salts which enter the duodenum 90-95% are reabsorbed from ileum in the portal vein and return to liver, to be excreted again. This is called enterohepatic circulation.
- The remaining 5-10% enter the colon and converted to salts of deoxycholic acid and lithocholic acid
- Enterohepatic circulation of bile salts is necessary because the limited pool of bile salts available for digestion and absorption of fats. The total pool of circulating bile salts is approx. 3.5gm. as an ordinary meal requires 6-8gm of bile salts to digest and absorbs fats. Therefore the total pool of bile salts must circulate twice during digestion of each meal.

FUNCTION OF BILE SALTS

- Emulsification of fats
- Acceleration of action of pancreatic lipase
- Micelle formation
- Absorption of fat soluble vitamins
- Cholorectic action - formation of bile cholesterol kept in soluble form in gall bladder
- Intestinal motility.

BILE PIGMENTS

- Mainly bilirubin and biliverdin
- Formed from haem portion of haemoglobin after destruction of RBCs in R-E system.
- Responsible for golden yellow colour of liver bile.

Other contents include -

- a. Phospholipid : primarily lecithin
- b. Cholesterol : by product of bile salt synthesis.

FUNCTION OF GALL BLADDER

1. Storage of bile : capacity of GB is 60ml. mucosa of GB actively absorbs fluid and electrolytes and concentrates them 5-6 times.
2. Control of flow of bile following meal
3. Regulates the alkalinity of the stored bile : absorption of HCO_3^- , Na and Cl, decreases the bile pH from 8.6 to 7.4
4. Regulates equalization of pressure in the biliary system.

	LIVER BILE	GALL BLADDER BILE
1. Colour	Light golden yellow	Almost black
2. Consistency	Almost water like	Thicker than the liver bile
3. pH	7.8 - 8.6	7.0 - 7.4
4. water	97%	89%
5. percentage of solids	2-4%	More ; 10-12%
6. bile salts / bile acids	120-180	More; 5-6 times the liver bile
7. bile pigments	50	50
8. electrolytes	-	Na^+ and K^+ increase by two times whereas Cl^- and HCO_3^- decrease by 5-6 times

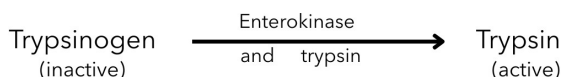
THE PANCREAS AND THE LIVER

- the pancreas is divided (from right to the left) into head, the neck, the body and tail.
 - The head is enlarged and lies within the concavity of the duodenum.
 - The tail reaches the hilum of the spleen
 - The entire organ lies posterior to the stomach separated from it by the lesser sac
 - The pancreas is a double function organ which contains both exocrine as well as endocrine cells.
- A. The exocrine part consists of secretory acini and duct cells that secrete pancreatic juice –
- a. The pancreatic juice is drained via two ducts
 - i. Ducts of wirsung
 - ii. Ducts of Santorini
- B. The endocrine part comprise of Islets of Langerhans which secretes hormone **insulin** and **glucagon**

COMPOSITION AND FUNCTION OF THE PANCREATIC JUICE

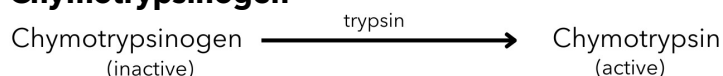
- 1.2 L - 1.5 L daily secretion
- pH 7.8 -8.4 markedly alkaline due to high HCO_3^-
- electrolytes- cations – Na^+ , K^+ , Ca^{2+} , magnesium, zinc
anions – HCO_3^- , Cl^- , HPO_4^{2-}
- enzymes –
 - pancreatic α amylase : digest starch and maltose
 - pancreatic lipase : hydrolyses fats to glycerol and fatty acids
 - pancreatic proteolytic enzymes secreted as inactive pro enzymes.

I. Trypsinogen



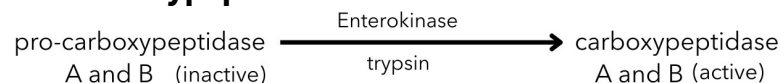
Hydrolyses protein to small polypeptides

II. Chymotrypsinogen



Chymotrypsinogen digests protein to small polypeptides.

III. Pro-carboxypeptidase A and B



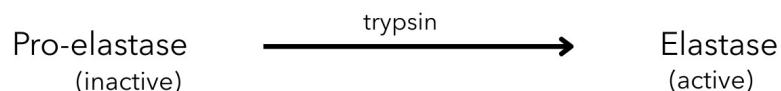
It splits peptide chain by the stepwise removal of amino acid residues from the free carboxyl group at the end of the chain

IV. Ribonuclease

V. Deoxyribonuclease

- they split nucleic acids of ribose and deoxyribose type into nucleotides.

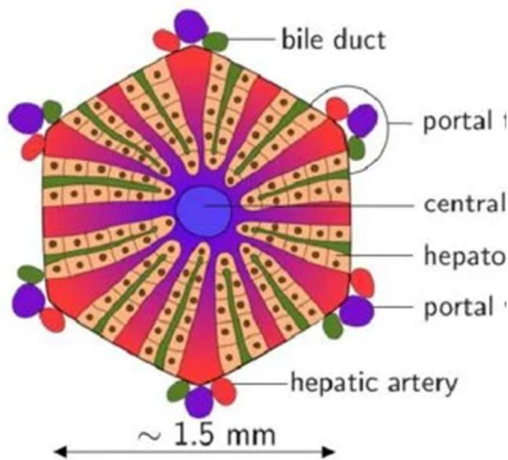
VI. Pro - elastase



It converts elastin and some other proteins into simplified substances

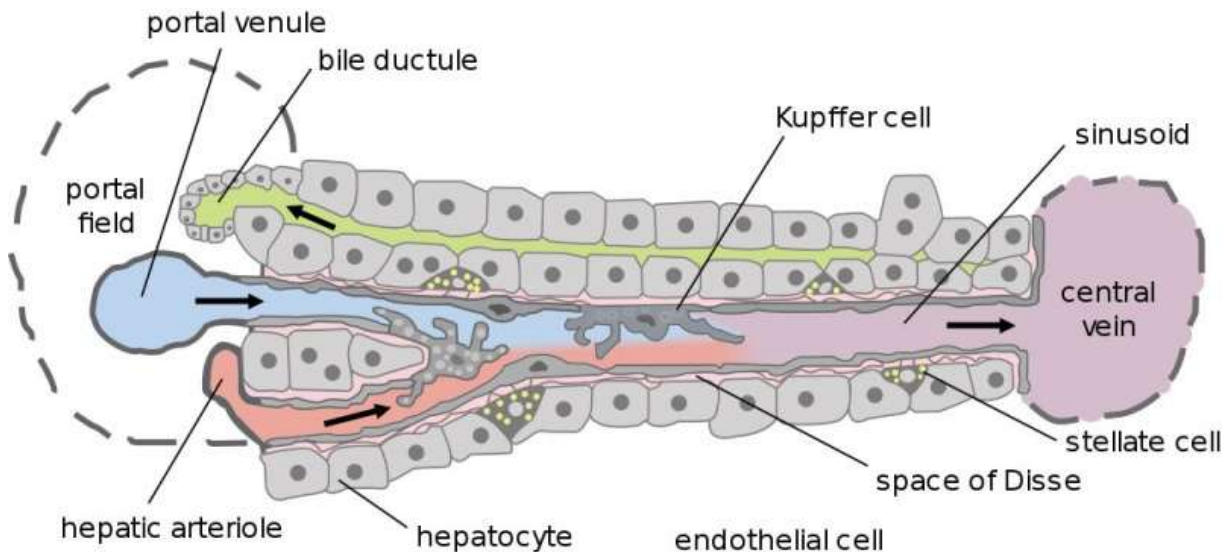
LIVER - PHYSIOLOGICAL ANATOMY

- Largest gland
- Weight - 1500 gm
- Anatomically - lobes - 2 (right and left)
- Each part has blood supply from hepatic artery, portal vein and own venous and biliary drainage.



STRUCTURAL CHARACTERISTICS

- Structural and functional units - hepatic lobules - about 1 lakh
- Each lobule has hepatocytes in columns
- Between columns of hepatocytes there present bile canaliculi
- Hepatocytes tunneled by blood sinusoids which drains into central vein.



- Sinusoids lined by endothelial cells, along with Kupffer cells in between.
- At periphery of each lobule present portal triad
- Contains - Portal vein, hepatic artery & bile duct.
- Acinus - is structural unit of liver

- Bile is formed in the hepatocytes and is discharged through canaliculi into intralobular bile ducts which joins to form right and left hepatic ducts. These ducts joins outside the liver to form common hepatic duct.

FUNCTION OF LIVER

1. **Secretory function** - secrete bile for digestion and absorption of fat
Defective fat digestion and absorption produces steatorrhoea (bulky, pale, greasy, foul smelling stool)
2. **Metabolic function-**
 - a. Carbohydrate metabolism
 - i. Glucogenesis
 - ii. Glycogenolysis
 - iii. Gluconeogenesis
 - b. Protein metabolism : liver synthesizes plasma proteins, blood clotting factors, enzymes, urea and lipoproteins from amino acids.
 - c. Fat metabolism : breakdown & synthesis of fat
 - i. Beta oxidation of fatty acids to form acetoacetic acid in mitochondria
 - ii. Synthesis of fatty acids from acetoacetic acid via krebs cycle in mitochondria
 - iii. Synthesis of cholesterol and phospholipid
3. **Synthesis function-**
 - a. Plasma proteins : specially albumin . globulin . it doesnot synthesize immunoglobulins, which are synthesized in R-E system
 - b. Clotting factors : I,II, V,VII,IX , X
 - c. Enzymes - SGOT, SGPT, alkaline phosphatase
 - d. Urea : liver removes ammonia to synthesize urea
 - e. Cholesterol : it is synthesized from acetoacetic acid
4. **Detoxicating and protective function -**
 - a. Kupffer cells removes bacteria and help immune mechanism
 - b. Detoxify drugs by oxidation, conjugation, reduction and hydrolysis
5. **Miscellaneous function -**
 - a. Storage function : Glucose in the form glycogen, vit B12, vit A
 - b. Hormone metabolism : inactivates insulin, glucagon, cortisol, estrogen and testosterone, aldosterone.
 - c. Site for formation and destruction of RBCs.